

Blueprint for the Lean Proof of the Order–Mass Identity, the Aldous–Broder Theorem, and the Stopped-Forest Law

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This document states the definitions, lemmas and proofs that the Lean development in `EAB/Paper/` follows, at the index types used in Lean. The docstrings of the Lean files cite its labels, such as `lem:orders` or `thm:stopped`.

The directed Aldous–Broder theorem is an existing result of Hu, Lyons and Tang, *A reverse Aldous–Broder algorithm* (2021), Theorem 3.4.¹ The formalization follows the manuscript’s proof through the order–mass identity.

1 Matrix and harmonic-extension notation

Throughout, V is a finite set with $n = |V| \geq 2$ and $\mathbf{L} \in \mathbb{R}^{V \times V}$. For subsets $R, C \subseteq V$, $\mathbf{L}[R, C]$ is the submatrix with rows R and columns C , and $\mathbf{L}[U] := \mathbf{L}[U, U]$. All matrix and vector indices are the indicated finite sets. We use the finite Leibniz formula for determinants also when the index set is empty. There is exactly one permutation of $\emptyset$, with sign 1, and its term is an empty product, equal to 1. Thus the determinant of the unique 0×0 matrix is 1. This value is needed only to read Theorem 4.1 when $|B| = 1$; the proof of Theorem 4.2 and Sections 5–7 do not use it. The vectors $\mathbf{1}_S, \mathbf{0}_S \in \mathbb{R}^S$ are constant, and $\mathbf{e}_u$ is the coordinate vector at u . For a nonempty finite set S and a matrix $A \in \mathbb{R}^{S \times S}$, write

$$\text{cof}_u(A) := \det(A[S \setminus \{u\}]), \quad \mathbf{d}(A) := (\text{cof}_u(A))_{u \in S}.$$

For $D \subseteq S$, the notation $\mathbf{1}_D \in \mathbb{R}^S$ means its indicator. Empty sums and products are 0 and 1.

Lemma 1.1 (Identity plus rank one). *For any finite set S and $a, y \in \mathbb{R}^S$,*

$$\det(\mathbf{I}_S + ay^\top) = 1 + y^\top a.$$

Proof. Expand the determinant by multilinearity in its columns. Terms using a in at least two columns vanish. The term using no such column is 1, while replacing column j of $\mathbf{I}_S$ by a has determinant a_j ; hence the remaining terms sum to $\sum_{j \in S} y_j a_j$. $\square$

Lemma 1.2 (Cofactors of a zero-row-sum matrix). *Let S be nonempty and $A \in \mathbb{R}^{S \times S}$ satisfy $A\mathbf{1}_S = \mathbf{0}_S$. Put $\mathbf{d} = \mathbf{d}(A)$. Then*

$$\text{adj}(A) = \mathbf{1}_S \mathbf{d}^\top, \quad \mathbf{d}^\top A = \mathbf{0}_S^\top.$$

For all $a, y \in \mathbb{R}^S$,

$$\det(A + ay^\top) = (y^\top \mathbf{1}_S)(a^\top \mathbf{d}). \tag{1}$$

¹<https://doi.org/10.1214/20-AIHP1101>

Proof. The nonzero vector $\mathbf{1}_S$ lies in $\ker A$, so $\det A = 0$. If $\text{rank } A = |S| - 1$, then $\ker A = \mathbb{R}\mathbf{1}_S$; the identity $A \text{adj}(A) = 0$ makes each column of $\text{adj}(A)$ constant, with value its diagonal entry $\text{cof}_u(A)$ in column u . If $\text{rank } A \leq |S| - 2$, every $(|S| - 1)$ -minor vanishes and both $\text{adj}(A)$ and $\mathbf{d}$ are zero. Thus in either case $\text{adj}(A) = \mathbf{1}_S \mathbf{d}^\top$. The identity $\text{adj}(A)A = \det(A)\mathbf{I}_S = 0$ gives $\mathbf{d}^\top A = 0$ by reading any row.

Multilinearity in columns, with terms that replace two columns by multiples of a vanishing, gives

$$\det(A + ay^\top) = \det A + \sum_{j \in S} y_j \det(A \text{ with column } j \text{ replaced by } a).$$

Expansion of a replaced column gives $\det(A \text{ with column } j \text{ replaced by } a) = \sum_{u \in S} a_u \text{adj}(A)_{ju} = a^\top \mathbf{d}$. Since $\det A = 0$, the displayed sum is $(\sum_j y_j)(a^\top \mathbf{d})$. $\square$

Definition 1.3 (Harmonic extension). *Let $U \subsetneq V$ and $\det \mathbf{L}[U] \neq 0$. For a boundary vector $m \in \mathbb{R}^{V \setminus U}$, define $\gamma^U(m) \in \mathbb{R}^V$ by*

$$\gamma^U(m)|_{V \setminus U} = m, \quad \gamma^U(m)|_U = -\mathbf{L}[U]^{-1} \mathbf{L}[U, V \setminus U]m. \quad (2)$$

When $U = \emptyset$, the second equation is empty and $\gamma^\emptyset(m) = m$. For $w \notin U$, abbreviate $\gamma^{U,w} := \gamma^U(\mathbf{e}_w)$.

Lemma 1.4 (Characterization, linearity, and unit boundary). *Under Definition 1.3, $\gamma^U(m)$ is the unique $g \in \mathbb{R}^V$ with $g|_{V \setminus U} = m$ and $(\mathbf{L}g)|_U = \mathbf{0}_U$. The map $m \mapsto \gamma^U(m)$ is linear. If in addition $\mathbf{L}\mathbf{1}_V = \mathbf{0}_V$, then*

$$\gamma^U(\mathbf{1}_{V \setminus U}) = \mathbf{1}_V, \quad \sum_{w \in V \setminus U} \gamma^{U,w} = \mathbf{1}_V. \quad (3)$$

Proof. The restriction of $\mathbf{L}\gamma^U(m)$ to U is $\mathbf{L}[U]\gamma^U(m)|_U + \mathbf{L}[U, V \setminus U]m = 0$. If two vectors have the required boundary and row values, their difference vanishes off U and its restriction to U is in the kernel of the invertible matrix $\mathbf{L}[U]$; hence they coincide. The formula in (2) is linear in m . Under zero row sums, $\mathbf{1}_V$ has boundary value $\mathbf{1}_{V \setminus U}$ and is annihilated on U , so uniqueness yields the first identity in (3). The second follows by linearity from $\mathbf{1}_{V \setminus U} = \sum_{w \notin U} \mathbf{e}_w$. $\square$

Lemma 1.5 (One-vertex harmonic update). *Let $U \subsetneq V$, $w \notin U$, and $U' = U \cup \{w\} \subsetneq V$. Assume $\det \mathbf{L}[U] \neq 0$ and $\det \mathbf{L}[U'] \neq 0$. For $m \in \mathbb{R}^{V \setminus U'}$, let $\tilde{m} \in \mathbb{R}^{V \setminus U}$ extend m by $\tilde{m}_w = 0$. Then*

$$\gamma^{U'}(m) = \gamma^U(\tilde{m}) + (\gamma^{U'}(m))_w \gamma^{U,w}. \quad (4)$$

Proof. Both sides are annihilated by the rows U of $\mathbf{L}$. They agree off U : at w the right side is $0 + (\gamma^{U'}(m))_w \cdot 1$; at every other vertex outside U its second term is zero and its first term equals m . Uniqueness in Lemma 1.4 proves the equality. $\square$

2 Aggregated matrices and one-vertex transfer

Definition 2.1 (Aggregation matrix). *For nonempty $U \subsetneq V$ with $\det \mathbf{L}[U] \neq 0$ and $M \in \mathbb{R}^{(V \setminus U) \times U}$, put*

$$\mathbf{N}(U, M) := \mathbf{I}_U + \mathbf{L}[U]^{-1} \mathbf{L}[U, V \setminus U]M \in \mathbb{R}^{U \times U}. \quad (5)$$

The matrix M has unit row sums if $M\mathbf{1}_U = \mathbf{1}_{V \setminus U}$.

Lemma 2.2 (Harmonic columns and zero row sums). *For $j \in U$, let $m_j = M_{\cdot j} \in \mathbb{R}^{V \setminus U}$. Then*

$$\mathbf{N}(U, M)_{\cdot j} = \mathbf{e}_j - \gamma^U(m_j)|_U. \quad (6)$$

If $\mathbf{L}\mathbf{1}_V = 0$ and M has unit row sums, then $\mathbf{N}(U, M)\mathbf{1}_U = 0$.

Proof. Equation (6) is (2) applied to the column m_j . Summing it over $j \in U$ and using linearity and $\sum_j m_j = M\mathbf{1}_U = \mathbf{1}_{V \setminus U}$ gives $\mathbf{N}(U, M)\mathbf{1}_U = \mathbf{1}_U - \gamma^U(\mathbf{1}_{V \setminus U})|_U = 0$ by (3). $\square$

Definition 2.3 (Fold). *Let $U \subsetneq V$ be nonempty, $w \in V \setminus U$, $U' = U \cup \{w\} \subsetneq V$, and $M' \in \mathbb{R}^{(V \setminus U') \times U'}$. For $\theta \in \mathbb{R}^U$ with $\sum_{j \in U} \theta_j = 1$, define its fold $M = \text{fold}_{w, \theta}(M') \in \mathbb{R}^{(V \setminus U) \times U}$ by*

$$M_{wj} = \theta_j, \quad M_{vj} = M'_{vj} + M'_{vw}\theta_j \quad (v \in V \setminus U', j \in U). \quad (7)$$

Write $s = M'_{\cdot w} \in \mathbb{R}^{V \setminus U'}$ and let $\tilde{s} \in \mathbb{R}^{V \setminus U}$ extend s by zero at w . For $j \in U'$, let $\tilde{m}'_j \in \mathbb{R}^{V \setminus U}$ similarly extend $M'_{\cdot j}$ by zero at w .

Lemma 2.4 (Fold identities). *If M' has unit row sums, so does M . For $j \in U$,*

$$\tilde{m}'_j = M_{\cdot j} - \theta_j(\tilde{s} + \mathbf{e}_w) \quad \text{in } \mathbb{R}^{V \setminus U}. \quad (8)$$

Proof. The row of M at w sums to $\sum_j \theta_j = 1$. For $v \in V \setminus U'$, its sum is $\sum_{j \in U} M'_{vj} + M'_{vw} \sum_{j \in U} \theta_j = \sum_{j \in U'} M'_{vj} = 1$. At w , the right side of (8) is $\theta_j - \theta_j = 0$; at $v \neq w$ it is $M'_{vj} + M'_{vw}\theta_j - \theta_j M'_{vw} = M'_{vj}$. $\square$

Lemma 2.5 (One-vertex transfer). *Assume $\mathbf{L}\mathbf{1}_V = 0$. Let $U \subseteq V$ be nonempty with $|U| \leq n - 2$, choose $w \in V \setminus U$, and put $U' = U \cup \{w\}$. Assume $\det \mathbf{L}[U] \neq 0$ and $\det \mathbf{L}[U'] \neq 0$. Let $M' \in \mathbb{R}^{(V \setminus U') \times U'}$ have unit row sums, let $\theta \in \mathbb{R}^U$ satisfy $\sum_{j \in U} \theta_j = 1$, and set $M = \text{fold}_{w, \theta}(M')$. With s and $\tilde{s}$ as in Definition 2.3,*

$$\sum_{x \in U'} \gamma_x^{U, w} \text{cof}_x(\mathbf{N}(U', M')) = \sum_{x \in U} \gamma^U(\tilde{s} + \mathbf{e}_w)_x \text{cof}_x(\mathbf{N}(U, M)). \quad (9)$$

In particular, the right side is independent of θ .

Proof. For $j \in U'$, let $m'_j = M'_{\cdot j}$ and define

$$\hat{G}_{xj} := \gamma^U(\tilde{m}'_j)_x \quad (x, j \in U'), \quad \rho_j := \gamma^{U'}(m'_j)_w, \quad a := \gamma^{U, w}|_{U'}.$$

Applying (4) to every column, followed by (6), gives

$$\mathbf{N}(U', M') = (\mathbf{I}_{U'} - \hat{G}) - a\rho^\top. \quad (10)$$

Since the rows of M' sum to one, linearity of harmonic extension and (3), read at w , give $\sum_{j \in U'} \rho_j = 1$. Also $\hat{G}_{wj} = \tilde{m}'_j(w) = 0$ for every j .

Put $D := \det(\mathbf{I}_{U'} - \hat{G})$. Equation (10), Lemma 2.2, and (1), applied to $A = \mathbf{N}(U', M')$, yield

$$D = \sum_{x \in U'} a_x \text{cof}_x(\mathbf{N}(U', M')). \quad (11)$$

The row w of $\mathbf{I}_{U'} - \hat{G}$ is the coordinate row $\mathbf{e}_w^\top$. Expanding its determinant along this row,

$$D = \det((\mathbf{I}_{U'} - \hat{G})[U]). \quad (12)$$

For $j \in U$, use (8), linearity of harmonic extension, and (6). They give

$$(\mathbf{I}_{U'} - \widehat{G})[U] = \mathbf{N}(U, M) + a' \theta^\top, \quad a' := \gamma^U(\tilde{s} + \mathbf{e}_w)|_U. \quad (13)$$

The fold has unit row sums by Lemma 2.4, so Lemma 2.2 gives $\mathbf{N}(U, M)\mathbf{1}_U = 0$. Applying (1) again and using $\sum_j \theta_j = 1$ yields

$$D = \sum_{x \in U} a'_x \text{cof}_x(\mathbf{N}(U, M)). \quad (14)$$

Equations (11) and (14) are (9). $\square$

Lemma 2.6 (Transfer onto a one-vertex base). *Assume the hypotheses of Lemma 2.5 on $\mathbf{L}$, U , w , U' and M' , with $U = \{r\}$. Then*

$$\sum_{x \in U'} \gamma_x^{U,w} \text{cof}_x(\mathbf{N}(U', M')) = \gamma^U(\tilde{s} + \mathbf{e}_w)_r. \quad (15)$$

Both cofactors on the left are determinants of 1×1 matrices.

Proof. Use the notation of the proof of Lemma 2.5. Equation (11) identifies the left side with D . Its derivation applies Lemma 1.2 to the 2×2 matrix $\mathbf{N}(U', M')$ only, and does not involve θ . By (12), D is the determinant of the 1×1 matrix $(\mathbf{I}_{U'} - \widehat{G})[U]$. The Leibniz formula on a one-element index set has the single term of the identity permutation, so D is the entry of that matrix at (r, r) . Let $\theta \in \mathbb{R}^U$ have its only coordinate equal to 1 and let $M = \text{fold}_{w,\theta}(M')$. By Lemmas 2.4 and 2.2, $\mathbf{N}(U, M)\mathbf{1}_U = 0$. The index set U has one element, so this says $\mathbf{N}(U, M)_{rr} = 0$. The entry at (r, r) of (13) is therefore $D = 0 + a'_r \theta_r = \gamma^U(\tilde{s} + \mathbf{e}_w)_r$. No determinant on an empty index set occurs. $\square$

3 Parent maps, root indicators, and growth orders

Let $B \subsetneq V$ be nonempty, put $Z = V \setminus B$, and let $k = |Z|$. A map $f : Z \rightarrow V$ is a *B-parent map* if, starting at each $v \in Z$ and repeatedly applying f while the current vertex lies in Z , the sequence reaches B . Let $q_B(v) \geq 1$ be its first hitting time of B , let $f^{(t)}(v)$ denote the iterates up to that time, and put $r_B(v) = f^{(q_B(v))}(v)$. The Lean implementation extends f to V by fixing every vertex of B , and uses this total map to define the iterates and the least positive hitting time. All iterates before that time lie in Z ; consequently $q_B(v)$, $r_B(v)$, descendant blocks, orders, and weights depend only on the original map $f : Z \rightarrow V$. The theorem statements accept this partial parent map; its extension is fixed by the definition and introduces no additional parent values.

Definition 3.1 (Root and descendant blocks). *For $j \in B$ and $u \in Z$, set*

$$Z_j := \{v \in Z : r_B(v) = j\}, \quad D_u := \{v \in Z : \text{for some } 0 \leq t < q_B(v), f^{(t)}(v) = u\}.$$

The root indicator $M_f \in \{0, 1\}^{Z \times B}$ is $(M_f)_{vj} = \mathbf{1}_{Z_j}(v)$. A vertex $w \in Z$ with $f(w) = i \in B$ is a child of i .

Lemma 3.2 (Root and child partitions). *The sets $(Z_j)_{j \in B}$ partition Z , so $M_f \mathbf{1}_B = \mathbf{1}_Z$. For each $i \in B$, the sets D_w over the children w of i are pairwise disjoint and partition Z_i .*

Proof. The first hit $r_B(v)$ exists and is unique for every $v \in Z$, giving the first partition. If $v \in Z_i$, the last non-root on its path before the first hit i is the unique $w = f^{(q_B(v)-1)}(v)$ with $f(w) = i$; thus $v \in D_w$. Conversely, if $v \in D_w$ and $f(w) = i \in B$, the path from v reaches w before any root and then reaches i , so $v \in Z_i$. The last non-root before the first root is unique, proving disjointness. $\square$

Definition 3.3 (Growth orders and weights). *A growth order of f is an ordering $\tau = (\tau_1, \dots, \tau_k)$ of Z such that*

$$U_l^\tau := B \cup \{\tau_1, \dots, \tau_l\} \quad (0 \leq l \leq k), \quad f(\tau_l) \in U_{l-1}^\tau \quad (1 \leq l \leq k).$$

Write $\mathcal{O}_B(f)$ for the set of these orders. Whenever the indicated harmonic extensions exist, set

$$\Pi_{B,f}(\tau) := \prod_{l=1}^{k-1} \gamma_{f(\tau_{l+1})}^{U_{l-1}^\tau, \tau_l}, \quad \mathcal{G}_i(B, f) := \sum_{\substack{\tau \in \mathcal{O}_B(f) \\ f(\tau_1) = i}} \Pi_{B,f}(\tau) \quad (i \in B).$$

For $k = 1$, the product is 1.

Lemma 3.4 (Existence and peeling of growth orders). *Every B -parent map has a growth order. Suppose $k \geq 2$. If $w \in Z$ is a child of $i \in B$, put $B' = B \cup \{w\}$ and let f' be the restriction of f to $V \setminus B'$. Then f' is a B' -parent map and*

$$\tau' \longmapsto (w, \tau')$$

is a bijection from $\mathcal{O}_{B'}(f')$ onto the growth orders of f whose first vertex is w . If $k \geq 2$ and $\tau' = (\tau'_1, \dots, \tau'_{k-1})$, then

$$\Pi_{B,f}(w, \tau') = \gamma_{f(\tau'_1)}^{B,w} \Pi_{B',f'}(\tau'). \quad (16)$$

Every proper prefix base of τ' is a proper prefix base of (w, τ') , with its prefix length increased by one.

Proof. If $k \geq 2$ and w is any child of a root, $B' = B \cup \{w\}$ remains proper. Every remaining parent path reaches B' , stopping at w if it meets w before B ; hence f' is a B' -parent map.

If $k = 1$, its only vertex w must satisfy $f(w) \in B$, since otherwise its path would never hit B ; hence (w) is a growth order. For $k \geq 2$, follow the path of any $v \in Z$ to its first root. Its predecessor $w \in Z$ is a child of that root. Induction on k , prepending w to an order for f' , proves existence.

For a fixed child w , its parent lies in B . After placing w first, the condition on each subsequent vertex is exactly the growth-order condition for f' and base B' , proving the bijection. The first factor of the product for (w, τ') is $\gamma_{f(\tau'_1)}^{B,w}$; shifting all remaining indices by one gives the factors of $\Pi_{B',f'}(\tau')$. The equality of the prefix bases is $B' \cup \{\tau'_1, \dots, \tau'_l\} = B \cup \{w, \tau'_1, \dots, \tau'_l\}$. $\square$

Lemma 3.5 (Peeling folds the root indicator). *Under the $k \geq 2$ hypotheses of Lemma 3.4, let $M' = M_{f'}$ and regard $\mathbf{e}_i \in \mathbb{R}^B$ as a vector with coordinate sum one. Then*

$$M_f = \text{fold}_{w, \mathbf{e}_i}(M'), \quad \tilde{s} + \mathbf{e}_w = \mathbf{1}_{D_w} \quad \text{in } \mathbb{R}^Z,$$

where $s = M'_{\cdot w}$ is the column of the new root w .

Proof. The row of M_f at w is $\mathbf{e}_i^\top$ because $f(w) = i$. For $v \neq w$, if its path first meets w before any vertex of B , its new root is w and its old root is i ; otherwise its new and old roots coincide in B . Consequently, for each $j \in B$, $(M_f)_{vj} = (M')_{vj} + (M')_{vw} \mathbf{1}_{\{i\}}(j)$, which is the fold formula. The column s indicates the remaining vertices whose paths first meet w ; adjoining w itself gives precisely D_w . $\square$

4 Order–mass identity

Theorem 4.1 (Order–mass identity). *Let $B \subsetneq V$ be nonempty, let $f : V \setminus B \rightarrow V$ be a B -parent map, and let $\mathbf{L} \in \mathbb{R}^{V \times V}$ satisfy $\mathbf{L}\mathbf{1}_V = 0$ and*

$$\det \mathbf{L}[U_l^\tau] \neq 0 \quad \text{for every } \tau \in \mathcal{O}_B(f) \text{ and } 0 \leq l \leq k-1. \quad (17)$$

Then, for every $i \in B$,

$$\mathcal{G}_i(B, f) = \text{cof}_i(\mathbf{N}(B, M_f)) = \det[(\mathbf{I}_B + \mathbf{L}[B]^{-1}\mathbf{L}[B, Z]M_f)[B \setminus \{i\}]]. \quad (18)$$

Proof. If $|B| = 1$, the right side of (18) is $\det(\emptyset) = 1$ by the Leibniz formula, and the identity is Theorem 4.2. The proof of that theorem uses the present one for a two-element root set only. Assume $|B| \geq 2$. First establish the assertion for $|B| \geq 2$ as a separate intermediate theorem. In that theorem, induct on $k = |Z|$, simultaneously for all proper root sets with at least two elements and their parent maps. The induction step enlarges the root set, so it stays within this class, and every determinant below is taken on a nonempty index set. Lemma 3.4 supplies a growth order, so (17) at $l = 0$ makes $\mathbf{N}(B, M_f)$ well defined.

If $k = 1$, write $Z = \{w\}$ and $i_0 = f(w) \in B$. The only growth order is (w) ; hence $\mathcal{G}_i(B, f) = \mathbf{1}_{\{i_0\}}(i)$. Zero row sums give $\mathbf{L}[B]\mathbf{1}_B + \mathbf{L}[B, \{w\}]\mathbf{1}_{\{w\}} = 0$, so $\mathbf{N}(B, M_f) = \mathbf{I}_B - \mathbf{1}_B \mathbf{e}_{i_0}^\top$. Its principal cofactor is 1 when $i = i_0$, because deleting that row and column leaves an identity matrix. When $i \neq i_0$, the remaining matrix is $\mathbf{I}_{B \setminus \{i\}} - \mathbf{1}_{B \setminus \{i\}} \mathbf{e}_{i_0}^\top$ and has determinant 0 by Lemma 1.1. This proves (18).

Suppose $k \geq 2$, fix $i \in B$, and abbreviate $N = \mathbf{N}(B, M_f)$ and $d_x = \text{cof}_x(N)$. Lemmas 3.2 and 2.2 imply $N\mathbf{1}_B = 0$; therefore Lemma 1.2 gives

$$\sum_{x \in B} d_x N_{xi} = 0. \quad (19)$$

Group the growth orders counted by $\mathcal{G}_i(B, f)$ according to their first vertex w , necessarily a child of i . For each such w , use the notation B', f', M' of Lemmas 3.4 and 3.5. Lemma 3.4 also carries the minor condition to f' : every proper prefix base of a new order is the corresponding proper prefix base of an old order beginning with w . In particular, $\det \mathbf{L}[B'] \neq 0$ because an order beginning with w exists. The inductive hypothesis at each $x \in B'$, combined with (16), gives

$$\mathcal{G}_i(B, f) = \sum_{\substack{w \in Z \\ f(w)=i}} \sum_{x \in B'} \gamma_x^{B, w} \text{cof}_x(\mathbf{N}(B', M')). \quad (20)$$

For each w , Lemma 3.5 identifies M_f as the fold of M' along $\mathbf{e}_i$ and identifies $\tilde{s} + \mathbf{e}_w = \mathbf{1}_{D_w}$. Lemma 2.5 changes the inner sum in (20) into $\sum_{x \in B} \gamma^B(\mathbf{1}_{D_w})_x d_x$. By Lemma 3.2, the D_w for the children of i partition Z_i . Linearity in Lemma 1.4 therefore gives

$$\mathcal{G}_i(B, f) = \sum_{x \in B} \gamma^B(\mathbf{1}_{Z_i})_x d_x.$$

The i th column of M_f is $\mathbf{1}_{Z_i}$, so (6) says $\gamma^B(\mathbf{1}_{Z_i})|_B = \mathbf{e}_i - N_{\cdot i}$. Using (19),

$$\mathcal{G}_i(B, f) = \sum_{x \in B} (\mathbf{e}_i - N_{\cdot i})_x d_x = d_i - \sum_{x \in B} d_x N_{xi} = d_i.$$

The definition of N gives the final displayed determinant in (18). $\square$

Theorem 4.2 (Single-root normalization). *Under the hypotheses of Theorem 4.1, if $B = \{r\}$, then*

$$\sum_{\tau \in \mathcal{O}_{\{r\}}(f)} \Pi_{\{r\},f}(\tau) = 1.$$

Proof. Put $B = \{r\}$, $Z = V \setminus \{r\}$ and $k = |Z| = n - 1$. The proof uses no determinant on an empty index set.

If $k = 1$, write $Z = \{w\}$. By Lemma 3.4 the only growth order is (w) , and $\Pi_{B,f}((w)) = 1$ by Definition 3.3. The sum is 1.

Suppose $k \geq 2$. Every growth order τ has $f(\tau_1) \in U_0^\tau = \{r\}$, so its first vertex is a child of r . Lemma 3.4 supplies a growth order; hence (17) at $l = 0$ gives $\det \mathbf{L}[B] \neq 0$. Fix a child w of r and use the notation $B' = \{r, w\}$, f' and $M' = M_{f'}$ of Lemmas 3.4 and 3.5. The bijection of Lemma 3.4 and (16), with the orders τ' grouped by $x = f(\tau'_1) \in B'$, give

$$\sum_{\substack{\tau \in \mathcal{O}_B(f) \\ \tau_1 = w}} \Pi_{B,f}(\tau) = \sum_{x \in B'} \gamma_x^{B,w} \mathcal{G}_x(B', f'). \quad (21)$$

By Lemma 3.4, f' is a B' -parent map, B' is proper because $k \geq 2$, the minor condition (17) holds for (B', f') , and $\det \mathbf{L}[B'] \neq 0$. Theorem 4.1 for the two-element root set B' gives $\mathcal{G}_x(B', f') = \text{cof}_x(\mathbf{N}(B', M'))$ for $x \in B'$; these cofactors are determinants of 1×1 matrices. That instance is proved by the induction on root sets with at least two elements, so the argument is not circular.

The matrix M' has unit row sums by Lemma 3.2, and $|B| = 1 \leq n - 2$. Lemma 2.6 with $U = B$ and $U' = B'$ evaluates the right side of (21) as $\gamma^B(\tilde{s} + \mathbf{e}_w)_r$, and Lemma 3.5 gives $\tilde{s} + \mathbf{e}_w = \mathbf{1}_{D_w}$. Hence

$$\sum_{\substack{\tau \in \mathcal{O}_B(f) \\ \tau_1 = w}} \Pi_{B,f}(\tau) = \gamma^B(\mathbf{1}_{D_w})_r.$$

Sum over the children w of r . By Lemma 3.2 the sets D_w partition Z_r , and $Z_r = Z$ because r is the only root. Linearity and (3) in Lemma 1.4 give

$$\sum_{\tau \in \mathcal{O}_B(f)} \Pi_{B,f}(\tau) = \gamma^B(\mathbf{1}_Z)_r = (\mathbf{1}_V)_r = 1.$$

Under the Leibniz value $\det(\emptyset) = 1$ this equality is Theorem 4.1 at $B = \{r\}$; the proof above does not rely on that reading. $\square$

5 Finite chains and confined excursions

Definition 5.1 (Chain, reversal, and first entrances). *A stochastic matrix $P \in \mathbb{R}^{V \times V}$ has $P_{uv} \geq 0$ and $P\mathbf{1}_V = \mathbf{1}_V$. It is irreducible if, for every $u, v \in V$, some $t \geq 0$ has $(P^t)_{uv} > 0$. A stationary distribution is a vector $\pi \in \mathbb{R}^V$ with $\pi_v > 0$, $\sum_v \pi_v = 1$, and $\pi^\top P = \pi^\top$. For every stochastic P put $\lambda_v = \det((\mathbf{I}_V - P)[V \setminus \{v\}])$ and $\Lambda = \sum_{v \in V} \lambda_v$. Given a stationary π , put*

$$D = \text{diag}(\pi), \quad P^* = D^{-1}P^\top D, \quad \mathbf{L} = \mathbf{I}_V - P^*. \quad (22)$$

Let $(X_t)_{t \geq 0}$ be the chain with transition matrix P started at r . Set $T_v = \inf\{t \geq 0 : X_t = v\}$ and $T_{\text{cov}} = \max_{v \in V} T_v$. On the event of finite cover, its first-entrance tree has edges (X_{T_v-1}, v) for $v \neq r$.

For $B \subsetneq V$ containing r , put $Z = V \setminus B$ and $T_B = \max_{v \in Z} T_v$. On the same event, its stopped first-entrance forest F_B has edges (X_{T_v-1}, v) for $v \in Z$. For a B -parent map f , write

$$w_P(f) := \prod_{v \in Z} \pi_{f(v)} P_{f(v)v}. \quad (23)$$

Lemma 5.2 (Proper confined minors and Green series). *For every irreducible stochastic P and every nonempty proper $U \subset V$,*

$$\det(\mathbf{I}_U - P[U]) > 0, \quad G_U := (\mathbf{I}_U - P[U])^{-1} = \sum_{t \geq 0} P[U]^t. \quad (24)$$

There is $0 \leq c_U < 1$ such that, for every $m \in \mathbb{N}$, every row sum of $P[U]^{m(n-1)}$ is at most c_U^m .

Proof. For each $i \in U$, irreducibility supplies a simple positive-probability path from i to $V \setminus U$ of length at most $n - 1$. Hence every row sum of $P[U]^{n-1}$ is less than one. Finiteness of U gives a common bound $0 \leq c_U < 1$, since row sums are nonnegative; submultiplicativity of the maximum row-sum norm proves the geometric estimate. The Neumann series converges and multiplies $\mathbf{I}_U - P[U]$ to $\mathbf{I}_U$. The same bound applies to $tP[U]$ for $0 \leq t \leq 1$, so $\det(\mathbf{I}_U - tP[U])$ never vanishes. It is continuous and equals 1 at $t = 0$; therefore it is positive at $t = 1$. $\square$

Lemma 5.3 (Stationary distribution from principal cofactors). *Every finite irreducible stochastic P has a unique stationary distribution. With λ and Λ as in Definition 5.1, it is $\pi = \lambda/\Lambda$; in particular $\lambda_v > 0$ and $\Lambda > 0$.*

Proof. The positivity of λ_v follows from Lemma 5.2 for $U = V \setminus \{v\}$; thus $\Lambda > 0$. Apply Lemma 1.2 to $\mathbf{I}_V - P$, whose row sums vanish. It gives $\lambda^\top (\mathbf{I}_V - P) = 0$, so $\pi = \lambda/\Lambda$ is a positive normalized stationary distribution. A nonzero $(n - 1)$ -minor makes $\text{rank}(\mathbf{I}_V - P) = n - 1$. Its left kernel is therefore one-dimensional, so normalization makes π unique. $\square$

Lemma 5.4 (Reversal and normalized inverse columns). *Let P be irreducible and stochastic, and let π be stationary. Then P^* is stochastic, $\mathbf{L}\mathbf{1}_V = 0$, and for every nonempty proper $U \subset V$,*

$$\det \mathbf{L}[U] = \det(\mathbf{I}_U - P[U]) > 0 \quad (25)$$

and

$$(G_U)_{ij} = \frac{\pi_j}{\pi_i} (\mathbf{L}[U]^{-1})_{ji} \quad (i, j \in U). \quad (26)$$

If $|U| \geq 2$ and $j \in U$,

$$(\mathbf{L}[U]^{-1})_{jj} = \frac{\det \mathbf{L}[U \setminus \{j\}]}{\det \mathbf{L}[U]}, \quad \frac{\mathbf{L}[U]^{-1} \mathbf{e}_j}{(\mathbf{L}[U]^{-1})_{jj}} = \gamma^{U \setminus \{j\}, j}|_U. \quad (27)$$

Proof. Stationarity gives $(P^* \mathbf{1}_V)_i = \pi_i^{-1} \sum_j P_{ji} \pi_j = 1$. Every entry $P_{ij}^* = \pi_j P_{ji} / \pi_i$ is nonnegative. Also $\mathbf{L}[U] = D[U]^{-1} (\mathbf{I}_U - P[U])^\top D[U]$; its determinant and inverse give the equal determinant and (26); the positivity in (25) follows from Lemma 5.2.

The first identity in (27) is Cramer's rule; its numerator is positive by Lemma 5.2, since $U \setminus \{j\}$ is nonempty and proper. Extend the inverse column $\mathbf{L}[U]^{-1} \mathbf{e}_j$ by zero off U and divide by its positive diagonal entry. It is one at j , zero outside U , and annihilated by the rows $U \setminus \{j\}$ of $\mathbf{L}$. Lemma 1.4 identifies it with the stated harmonic extension. $\square$

Lemma 5.5 (Finite cover and first-entrance forest). *For an irreducible finite chain, every T_v and T_{cov} is finite almost surely. For a nonempty proper $B \subset V$ with $r \in B$, the first-entrance edges of F_B form a B -parent map, and the first-entrance tree is an arborescence rooted at r .*

Proof. Apply the uniform escape estimate of Lemma 5.2 to $U = V \setminus \{v\}$. The chance of avoiding v for $m(n-1)$ further steps decays geometrically, from any current state outside v . Thus $T_v < \infty$ almost surely; finiteness of V gives the cover claim. If $v \notin B$ is entered from u , then $T_u < T_v$. Following its parent edges therefore strictly decreases first entrance times, so it cannot cycle and must end at a vertex of B . With $B = \{r\}$ this is an arborescence. $\square$

Lemma 5.6 (Confined-excursion formula). *Under Definition 5.1, let P be irreducible, let $B \subsetneq V$ contain r , and let f be a B -parent map. For a growth order $\tau = (\tau_1, \dots, \tau_k)$ put $\tau_0 = r$ and $U_l = U_l^\tau$. Then*

$$\mathbb{P}_r(F_B = f, \text{ discovery order} = \tau) = \left(\prod_{v \in Z} P_{f(v)v} \right) \prod_{l=1}^k (G_{U_{l-1}})_{\tau_{l-1}, f(\tau_l)}. \quad (28)$$

For an ordering that is not a growth order, the probability is zero.

Proof. Immediately before the first entry into τ_l , the visited non-roots are $\tau_1, \dots, \tau_{l-1}$, while the walk may visit any root in B . It starts this interval at τ_{l-1} (at r if $l = 1$), spends $t_l \geq 0$ steps within U_{l-1} , then takes the edge $f(\tau_l) \rightarrow \tau_l$. For fixed t_l this interval has weight $(P[U_{l-1}]^{t_l})_{\tau_{l-1}, f(\tau_l)} P_{f(\tau_l), \tau_l}$. On finite cover, the successive intervals and their lengths are uniquely determined; conversely, their concatenation gives exactly the stated event. The cases indexed by the length vector are disjoint. The Markov property makes their weights multiply, and nonnegativity permits summing over all length vectors as a product of the separate sums. Summing each length with (24) gives (28). Every recorded parent has already been visited, so an incompatible ordering has probability zero. $\square$

Lemma 5.7 (Reversal and determinant telescoping). *Under Lemma 5.6, let π be the stationary distribution of P and use the forest weight (23). Then for each $\tau \in \mathcal{O}_B(f)$,*

$$\mathbb{P}_r(F_B = f, \text{ discovery order} = \tau) = \frac{\det \mathbf{L}[B]}{\pi_r(\prod_{v \in Z} \pi_v) \Lambda} w_P(f) (\mathbf{L}[B]^{-1})_{f(\tau_1), r} \Pi_{B, f}(\tau). \quad (29)$$

The formula includes $k = 1$, when $\Pi_{B, f}(\tau) = 1$.

Proof. Apply (26) to each Green entry of (28). Since $\prod_{l=1}^k \pi_{\tau_{l-1}} = \pi_r(\prod_{v \in Z} \pi_v)/\pi_{\tau_k}$, the factors outside the inverse entries become $w_P(f) \pi_{\tau_k} / [\pi_r \prod_{v \in Z} \pi_v]$.

For $2 \leq l \leq k$ apply (27) to $U_{l-1} = U_{l-2} \cup \{\tau_{l-1}\}$ and $j = \tau_{l-1}$. It gives

$$(\mathbf{L}[U_{l-1}]^{-1})_{f(\tau_l), \tau_{l-1}} = \gamma_{f(\tau_l)}^{U_{l-2}, \tau_{l-1}} \frac{\det \mathbf{L}[U_{l-2}]}{\det \mathbf{L}[U_{l-1}]}.$$

The first inverse entry remains $(\mathbf{L}[B]^{-1})_{f(\tau_1), r}$. The other determinant ratios telescope to $\det \mathbf{L}[B] / \det \mathbf{L}[U_{k-1}]$; this quotient is 1 when $k = 1$. Since $U_{k-1} = V \setminus \{\tau_k\}$, (25) and Lemma 5.3 give $\det \mathbf{L}[U_{k-1}] = \lambda_{\tau_k} = \pi_{\tau_k} \Lambda$. The remaining harmonic factors form $\Pi_{B, f}(\tau)$, proving (29). $\square$

6 Aldous–Broder

Definition 6.1 (Tree and forest families). *For nonempty $B \subseteq V$, let $\mathcal{F}_B$ be the finite set of B -parent maps, identified with their edge sets $\{(f(v), v) : v \in V \setminus B\}$. If $B = \{r\}$, these are the arborescences rooted at r .*

Theorem 6.2 (Aldous–Broder for an irreducible chain). *Let P be irreducible and stochastic, let π be its stationary distribution, and start the covering walk at $r \in V$. For every $f \in \mathcal{F}_{\{r\}}$, its first-entrance tree F satisfies*

$$\mathbb{P}_r(F = f) = \frac{\prod_{v \neq r} \pi_{f(v)} P_{f(v)v}}{\Lambda \prod_{v \in V} \pi_v} = \frac{\prod_{v \neq r} \pi_{f(v)} P_{f(v)v}}{\sum_{f' \in \mathcal{F}_{\{r\}}} \prod_{v \neq r} \pi_{f'(v)} P_{f'(v)v}}. \quad (30)$$

Proof. In Lemma 5.7 take $B = \{r\}$. Every growth order starts with a vertex whose parent is r , and the one-by-one inverse entry is $(\mathbf{L}[\{r\}]^{-1})_{rr} = 1/\det \mathbf{L}[\{r\}]$. Thus, for each compatible discovery order τ ,

$$\mathbb{P}_r(F = f, \text{ order} = \tau) = \frac{w_P(f)}{\Lambda \prod_{v \in V} \pi_v} \Pi_{\{r\}, f}(\tau).$$

Summing over growth orders and applying Theorem 4.2 gives the first expression in (30). By Lemma 5.5, the first-entrance tree belongs to exactly one member of $\mathcal{F}_{\{r\}}$ almost surely. Sum the first expression over this finite family; the resulting equality $1 = \sum_{f'} w_P(f')/(\Lambda \prod_v \pi_v)$ identifies its denominator with the tree-weight sum, giving the second expression. $\square$

7 Stopped multi-root forest law

Definition 7.1 (Order masses and completion factor). *Let P be irreducible and stochastic with stationary distribution π , let $B \subsetneq V$ be nonempty, $r \in B$, and $f \in \mathcal{F}_B$. With the matrix $\mathbf{L}$ in (22), write*

$$g(B, f) := (\mathcal{G}_i(B, f))_{i \in B}, \quad \varphi_r(f) := g(B, f)^\top \mathbf{L}[B]^{-1} \mathbf{e}_r.$$

Lemma 7.2 (Forest as a marginal of the covering tree). *Let P be irreducible and stochastic with stationary distribution π , let $B \subsetneq V$ be nonempty, choose $r \in B$, and let $f \in \mathcal{F}_B$. Define*

$$\mathcal{C}_r(f) := \{h \in \mathcal{F}_{\{r\}} : h|_{V \setminus B} = f\}.$$

Then

$$\mathbb{P}_r(F_B = f) = \sum_{h \in \mathcal{C}_r(f)} \mathbb{P}_r(F = h) = \frac{\sum_{h \in \mathcal{C}_r(f)} w_P(h)}{\Lambda \prod_{v \in V} \pi_v}. \quad (31)$$

For $h \in \mathcal{C}_r(f)$, $w_P(h) = w_P(f) \prod_{j \in B \setminus \{r\}} \pi_{h(j)} P_{h(j)j}$.

Proof. The first-entrance edge into any $v \in Z$ is fixed at time $T_v \leq T_B \leq T_{\text{cov}}$. Thus the stopped forest is exactly the restriction of the full first-entrance tree to edges entering Z . The disjoint events $\{F = h\}$ with $h \in \mathcal{C}_r(f)$ partition $\{F_B = f\}$. Apply Theorem 6.2 to each event. The factorization of $w_P(h)$ separates its edges entering Z from those entering $B \setminus \{r\}$. $\square$

Theorem 7.3 (Stopped forest law and completion weights). *Let P be irreducible and stochastic with stationary distribution π , let $B \subsetneq V$ be nonempty, choose a seed $r \in B$, and let $f \in \mathcal{F}_B$. Then*

$$\mathbb{P}_r(F_B = f) = \frac{\det \mathbf{L}[B]}{\pi_r(\prod_{v \in Z} \pi_v) \Lambda} w_P(f) \varphi_r(f). \quad (32)$$

The total weight of the arborescence completions of f is

$$\sum_{h \in \mathcal{C}_r(f)} \prod_{j \in B \setminus \{r\}} \pi_{h(j)} P_{h(j)j} = \det \mathbf{L}[B] \left(\prod_{j \in B \setminus \{r\}} \pi_j \right) \varphi_r(f). \quad (33)$$

Proof. Sum (29) over the compatible growth orders, grouping them by the root $i = f(\tau_1)$. The common factor is $\det \mathbf{L}[B] w_P(f) / [\pi_r(\prod_{v \in Z} \pi_v) \Lambda]$; the remaining sum is

$$\sum_{i \in B} \mathcal{G}_i(B, f) (\mathbf{L}[B]^{-1})_{ir} = g(B, f)^\top \mathbf{L}[B]^{-1} \mathbf{e}_r = \varphi_r(f).$$

This proves (32), including when $w_P(f) = 0$.

If $w_P(f) > 0$, compare (32) with (31), use the factorization of $w_P(h)$, and cancel $w_P(f)$. Since $\prod_{v \in V} \pi_v = (\prod_{v \in Z} \pi_v)(\prod_{j \in B} \pi_j)$, the resulting equality is (33).

For arbitrary f , set $P_\varepsilon = (1 - \varepsilon)P + \varepsilon \mathbf{1}_V \mathbf{1}_V^\top / n$, with $0 < \varepsilon < 1$. All its entries and hence all its forest weights are positive, so (33) holds for P_ε . By Lemma 5.3, its stationary distribution is the normalized cofactor vector of $\mathbf{I}_V - P_\varepsilon$, and therefore converges to π as $\varepsilon \downarrow 0$. Lemma 5.4 ensures that all principal minors used in the harmonic extensions and in $\mathbf{L}[B]^{-1}$ are positive at the limit; continuity keeps them nonzero for sufficiently small ε . The set of growth orders is finite, so its order masses and $\varphi_r(f)$ are continuous at the limit. Every term on both sides of (33) is thus continuous there. Taking the limit proves the formula also when $w_P(f) = 0$. $\square$

8 Mathematical contract for the Lean development

The three public endpoints are Theorems 4.1, 6.2, and 7.3, including both displayed conclusions of Theorem 7.3. Theorem 4.2 and the lemmas before each endpoint are its proof obligations, not additional application targets. In Sections 1–4, $\mathbf{L}$ is an arbitrary real zero-row-sum matrix where stated. In the chain sections it is specialized to $\mathbf{I} - P^*$ by (22).

The following boundary conditions are part of the formalized statements.

1. A root set in the order–mass theorem is nonempty and proper, so $k = |V \setminus B| \geq 1$. The peeling bijection and the fold are used only for $k \geq 2$; the $k = 1$ case is proved directly. No “growth order” for $B = V$ is silently used.
2. Theorem 4.1 needs nonzero minors only for proper bases U_l^τ of growth orders of the *given* parent map. Neither all intermediate subsets nor positivity of the matrix entries is assumed. The conclusion holds for every root, including one with no children. Only its reading at $|B| = 1$ takes a cofactor of a 1×1 matrix, with the Leibniz value $\det(\emptyset) = 1$. Theorem 4.2 is proved through Lemma 2.6 and partition of unity, not by that reading; the Aldous–Broder and stopped-forest endpoints depend on it only through Theorem 4.2.

3. The Aldous–Broder and stopped-forest statements assume an irreducible row-stochastic matrix, not reversibility, aperiodicity, or strictly positive transitions. Their probabilities refer to the Markov chain started at the stated seed. The first-entrance edge of a non-root v is (X_{T_v-1}, v) ; the seed r is never assigned such an edge.
4. Equation (33) is asserted even when $w_P(f) = 0$. It cannot be obtained by cancelling that weight. Its separate perturbation-and-continuity argument is required.

The manuscript’s B -parent map is a function on $Z = V \setminus B$, as is the Lean type `ParentAssignment B`. The public theorems accept this partial map; `extendParents` fixes the roots only to express its iterates. Likewise, $\text{cof}_i(A)$ always deletes both row and column i ; the two matrix levels $\mathbf{L}[B]$ and $\mathbf{N}(B, M_f)$ must not be conflated.

The notation for completion weights differs between the manuscript and this blueprint. Here $\varphi_r(f)$ denotes the algebraic factor $g(B, f)^\top \mathbf{L}[B]^{-1} \mathbf{e}_r$, represented by `Chain.completionFactor`. The manuscript’s completion weight is the sum on the left of (33), including the additional factor $\det \mathbf{L}[B] \prod_{j \in B \setminus \{r\}} \pi_j$ on the right.